

ON THE NONCENTRAL DISTRIBUTION OF A CERTAIN CORRELATION MATRIX

D.G. Kabe

St. Mary's University, Halifax, N.S., Canada

SUMMARY

Let A and B be two $p \times p$ positive definite symmetric random matrices, A Wishart distributed, and B noncentral Wishart distributed. Then a procedure is given for explicitly evaluating the density of the correlation matrix $R = Q(A+B)^{-1/2} B(A+B)^{-1/2} Q'$, where Q is an orthogonal matrix.

1. Introduction

Let A and B be two $p \times p$ positive definite symmetric random matrices having the densities

$$(1) \quad g(A) = K \exp(-\frac{1}{2} \text{tr } A) |A|^{\frac{1}{2}(N-r-p-1)},$$

and

$$(2) \quad g(B) = K \exp(-\frac{1}{2} \text{tr } B) |B|^{\frac{1}{2}(r-p-1)} {}_0F_1[\frac{1}{2}r, \frac{1}{2} \Omega B],$$

where the hypergeometric series of matrix argument ΩB is defined by Constantine (1963), p. 1276. Then Constantine (1963), p. 1280, and recently Troskie (1969), p. 113, equation (21), define a certain correlation matrix R by the relations

$$(3) \quad B = G^{1/2} R G^{1/2}, \quad G = A + B.$$

Both Constantine and Troskie obtain integral representations for the marginal density of R . The density of QRQ' when Ω has rank two is given by Kabe (1963), and by Hayakawa (1967), p. 12, equation 38, when Ω is of a general rank. However, Hayakawa's claim that his expression is the density of R is wrong. Hayakawa obtains the density of QRQ' , where Q is an arbitrary orthogonal matrix of rank p . Hayakawa does not supply a derivation of this important result, and perhaps, a derivation on record might be useful.

All the densities occurring in this paper are derived without the normalizing constants and K denotes these constants. We assume that all integrals occurring in this paper are evaluated over the appropriate ranges of their variables of integration.

2. Distribution of R

We write the joint density of A and $B = YY'$, Y is $p \times r$, $r \geq p$, as

$$(4) \quad g(A, Y) = K \exp \left\{ -\frac{1}{2} \text{tr} (A + YY') + \text{tr} \mu Y' \right\} |A|^{\frac{1}{2}(N-r-p-1)}$$

where $\text{pxr } \mu$ is of rank p (for convenience), and then setting $A + YY' = G$, $Y = G^{\frac{1}{2}}Z$, $ZZ' = R$, we find the joint density of G and R to be

$$(5) \quad g(G, R) = K \exp \left\{ -\frac{1}{2} \text{tr} G \right\} |G|^{\frac{1}{2}(N-r)} \\ |G|^{\frac{1}{2}(r-p-1)} |I-R|^{\frac{1}{2}(N-r-p-1)} |G|^{\frac{1}{2}} \mu \mu' G^{\frac{1}{2}} \cdot p/2 \\ \int |R-V(G^{\frac{1}{2}} \mu \mu' G^{\frac{1}{2}})^{-1} V'|^{-\frac{1}{2}(r-2p-1)} \exp \left\{ \text{tr} V' \right\} dV,$$

where V is a $p \times p$ matrix, the derivation (5) is simple and is obtained by using generalized Suerdrup's lemma, see Kabe (1965), or Anderson's (1946) integral representation of the noncentral Wishart density. The density (5) may also be written as

$$(6) \quad g(G, R) = K \exp \left\{ -\frac{1}{2} \text{tr} G \right\} |G|^{\frac{1}{2}(N-r)} |G|^{\frac{1}{2}(r-p-1)} \\ |I - QRQ'|^{-\frac{1}{2}(N-r-p-1)} \\ \int |QRQ' - QVV'Q'|^{-\frac{1}{2}(r-2p-1)} \exp \left\{ \text{tr} \Lambda QV \right\} dV,$$

where Λ^2 is the diagonal matrix of the roots of

$$(7) \quad |\mu \mu' - \lambda G^{-1}| = 0,$$

and Q is an arbitrary orthogonal matrix. Denoting QRQ' by R itself and setting $QV = Z$, we find that

$$(8) \quad g(G, R) = K \exp \left\{ -\frac{1}{2} \text{tr} G \right\} |G|^{\frac{1}{2}(N-r)} |G|^{\frac{1}{2}(r-p-1)} \\ |I-R|^{\frac{1}{2}(N-r-p-1)} |R|^{\frac{1}{2}(r-2p-1)} \\ \int |I-Z'R^{-1}Z|^{-\frac{1}{2}(r-2p-1)} \exp \left\{ \text{tr} \Lambda Z \right\} dZ.$$

the density of R is found from (8) by integrating (8) with respect to G and Z . However, the required integral is the $\frac{1}{2}(N-r)$ -th moment of $|G|$, and using a result of Constantine's ((1963), p. 1279, equation 39), see also Kabe (1963), equation (3.6) to (3.12), we find that

$$(9) \quad g(R) = K |R|^{\frac{1}{2}(r-p-1)} |I-R|^{\frac{1}{2}(N-r-p-1)} {}_1F_1 \left[\frac{1}{2}N, \frac{1}{2}r, \frac{1}{2} \Omega R \right],$$

where $R = Q(A+B)^{-\frac{1}{2}} B(A+B)^{-\frac{1}{2}} Q'$.

Incidentally, we must mention that it is not possible (or it is difficult) to obtain the density of R , defined by (3), even in the linear case, although both Constantine (1963), p. 1280, and Troskie (1969), p. 115, confirm that this is possible. In this case, the noncentral part will involve the root of

$$(10) \quad |G^{\frac{1}{2}} \delta \delta' G^{\frac{1}{2}} - \lambda R^{-1}| = 0,$$

where $\delta' = (\mu, 0, 0, \dots, 0)$ and μ^2 is the only root of $\mu\mu'$, as $\mu\mu'$ now is of rank unity. How does one integrate with respect to G in this case? The only possibility to readily integrate with respect to G in this case is to multiply both sides of (10) by $|Q|$, where Q is an orthogonal matrix that transforms $\delta' G^{\frac{1}{2}}$ to the vector $((\delta' G \delta)^{\frac{1}{2}}, 0, 0, \dots, 0)$. This procedure reduces (10) to the equation

$$(11) \quad \left| \begin{pmatrix} \delta' G \delta & 0 \\ 0 & 0 \end{pmatrix} - \lambda Q R^{-1} Q' \right| = 0,$$

and this is the procedure we have followed in the general case.

It may be noted that the density of QRQ' where R is defined by

$$(12) \quad A = CRC', \quad A + B = CC',$$

C a lower triangular matrix is also given by (9), see Kshirsagar (1961).

REFERENCES

- ANDERSON, R.W. (1946). The noncentral Wishart distribution and certain problems of multivariate statistics. *Ann. Math. Statist.* **17**, 409-431.
- CONSTANTINE, A.G. (1963). Some noncentral distribution problems in multivariate analysis, *Ann. Math. Statist.* **34**, 1270-1285.
- HAYAKAWA, TAKESI (1967). On the distribution of the maximum latent root of a positive definite symmetric random matrix. *Ann. Inst. Stat. Math.* **19**, 1-17.
- KABE, D.G. (1963). Some results on the distribution of two random matrices used in classification procedures. *Ann. Math. Statist.* **34**, 181-185.
- KABE, D.G. (1965). Generalization of Sverdrup's lemma and its applications to multivariate distribution theory. *Ann. Math. Statist.* **36**, 671-676.
- KSHIRSAGAR, A.M. (1961). The noncentral multivariate beta distribution. *Ann. Math. Statist.* **32**, 104-111.
- TROSKIE, C.G. (1969). The generalized multiple correlation matrix. *S. Afr. Statist. J.*, **3**, 109-121.

Manuscript received 1st February 1971.